

Expt.No: 67
Date:

Measurement of Earth Resistance

Aim:

To measure the earth resistance of an electrical equipment.

Apparatus Required:

S.No	Name of the Apparatus	Range/type	Quantity
1.	Megger		
2.	Connecting Rods		
3.	Connecting Wires		
4.			

Theory:

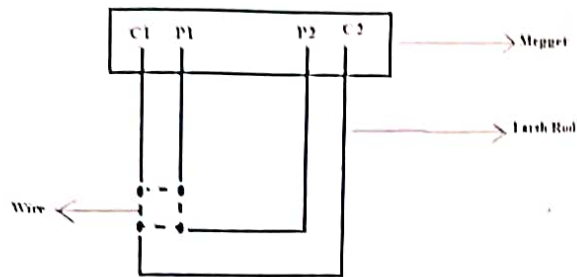
For this experiment we have to use the Megger. It is an instrument for testing the insulation resistance of the order of mega ohms.

A megger consists of an emf source and a voltmeter. The voltmeter scale is calibrated in ohms. In measurement, the emf of the self-contained source should be equal that of the source used in calibration. The deflection of the moving system depends on the ratio of the currents in the coils and is independent of the applied voltage. The value of unknown resistance can be found directly from the scale of the instrument. Figure shows detailed diagram of a megger. It consists of a hand driven dc generator of emf about 500V. The permanent dc meter has two moving coils. First one is deflecting coil and another one is controlling coil. The deflecting coil is connected to the generator through a resistor R_2 . The torque due to two coils opposes each other. It consists of three terminals E (earth terminal) and L (line terminal) and G (guard wire terminal).

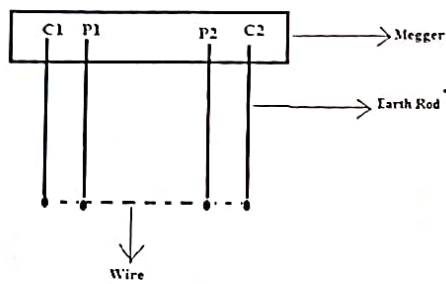
Operation:

When the terminals are open circuited, no current flows through the deflecting coil. The torque to the controlling coil moves the pointer to one end of the scale. When the terminals are short circuited, the torque due to the controlling coil moves the pointer to the other end of the scale i.e. zero mark. In between the two extreme positions the scale is calibrated to indicate the value of unknown resistance directly. The unknown insulation resistance is the combination of insulation volume resistance and surface leakage resistance. The guard wire terminal makes the surface leakage current to bypass the instrument hence only insulation resistance is measured.

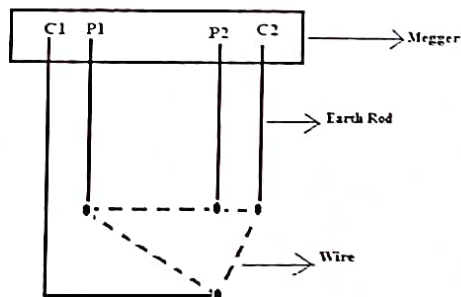
SQUARE POSITION



HORIZONTAL POSITION



TRIANGULAR POSITION



Tabulation:

S.No	Position (Distance in Meter)	Resistance (Ω)
1	Square C1-P1: ___ m, C1-P2: ___ m, C1-C2: ___ m, P1-P2: ___ m, P1-C2: ___ m, P2-C2: ___ m	
2	Horizontal C1-P1: ___ m, C1-P2: ___ m, C1-C2: ___ m, P1-P2: ___ m, P1-C2: ___ m, P2-C2: ___ m	
3	Triangular C1-P1: ___ m, C1-P2: ___ m, C1-C2: ___ m, P1-P2: ___ m, P1-C2: ___ m, P2-C2: ___ m	

Result: